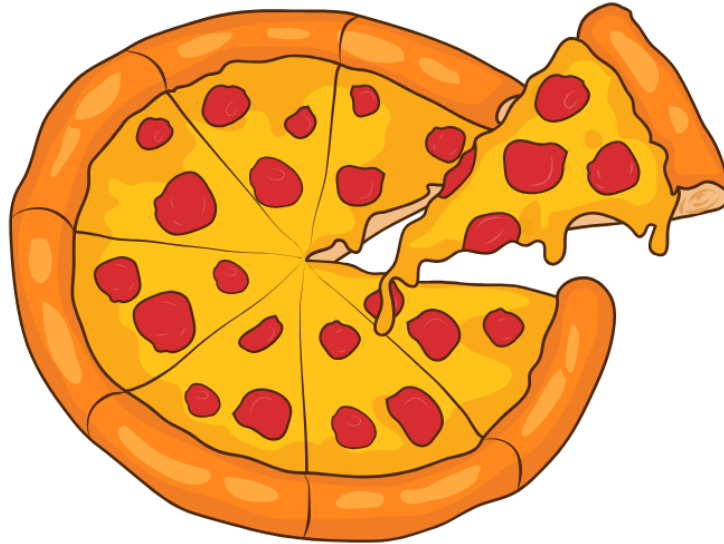




Pizza Sales Analysis

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PIZZA SALES ANALYSIS

Project Overview

This project focuses on analyzing pizza sales data to gain valuable business insights using SQL Server, Excel, and Power BI.

The dataset contains transactional records of pizza orders, including details about order dates, pizza categories, sizes, quantities, total prices, and ingredient usage.

The primary goal is to identify sales patterns, top-performing pizzas, customer preferences, and optimize ingredient procurement. Insights are visualized through an interactive Power BI dashboard to support decision-making for business growth.

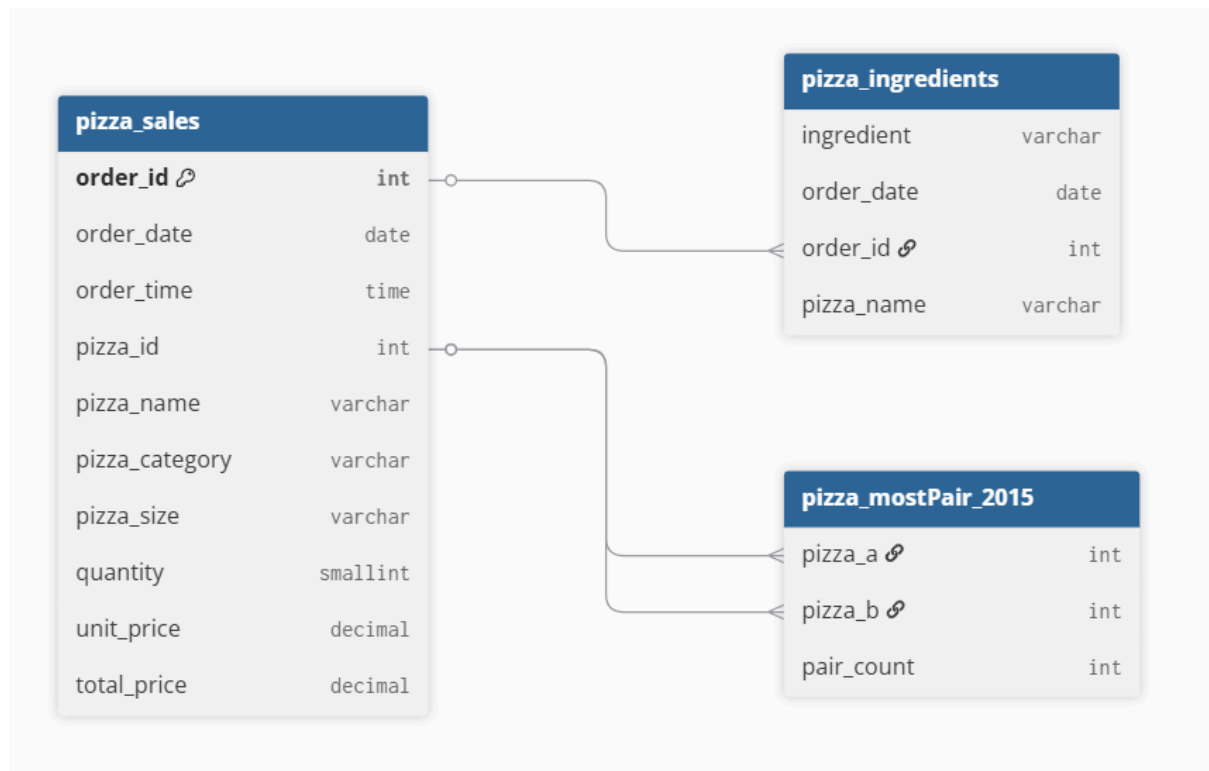
Tools and Technologies

- SQL Server Management Studio -> Data cleaning, transformation, and querying
- Power BI -> Transformation and Interactive dashboard creation and visualization
- Excel -> Source file and initial data preprocessing

Project Highlights

- Analyzed 21K total orders with 50K pizzas sold generating \$817.86K revenue.
- Discovered peak sales on Fridays at lunchtime (12:00–15:00).
- Identified July and May as the highest sales months.
- Found that Large pizzas contribute ~46% of revenue, while XXL pizzas contribute the least.
- Classic & Supreme pizzas dominate sales (~52% of total revenue).
- Top sellers: Thai Chicken, BBQ Chicken, California Chicken, Classic Deluxe, Spicy Italian.
- Ingredient analysis revealed Garlic, Tomatoes, Red Onions, and Mozzarella as the most used ingredients, while niche items like Brie Carre Cheese, Thyme, and Prosciutto had minimal usage.
- Provided procurement recommendations to optimize inventory management and reduce spoilage.

ERD



Pizza Sales SQL Queries

1. Total Revenue

```
select sum(total_price) as total_revenue from pizza_sales
```

	total_revenue
1	817860.049999993

2. Average Order Value

```
select sum(total_price) / count(distinct order_id) as avg_order_value from pizza_sales
```

	avg_order_value
1	38.3072622950816

3. Total Pizzas Sold

```
select sum(quantity) as total_pizzas_sold from pizza_sales
```

	total_pizzas_sold
1	49574

4. Total Orders

```
select count(distinct order_id) as total_orders from pizza_sales
```

	total_orders
1	21350

5. Average Pizza per Order

```
select cast(cast(sum(quantity) as decimal(10,2)) /  
cast(count(distinct order_id) as decimal(10,2)) as decimal(10,2)) as avg_pizza_perOrder from pizza_sales
```

avg_pizza_perOrder
2.32

6. Hourly Trend for Total Orders

```
select  
datepart(hour, order_time) as order_hours,  
count(distinct order_id) as total_orders  
from pizza_sales  
group by datepart(hour, order_time)  
order by total_orders desc
```

	order_hours	total_orders
1	12	2520
2	13	2455
3	18	2399
4	17	2336
5	19	2009
6	16	1920
7	20	1642
8	14	1472
9	15	1468
10	11	1231
11	21	1198
12	22	663
13	23	28
14	10	8
15	9	1

7. Daily Trend for Total Orders

```
select DATENAME(DW, order_date) as order_day, COUNT(distinct order_id) as total_orders from pizza_sales  
group by DATENAME(DW, order_date)
```

	order_day	total_orders
1	Saturday	3158
2	Wednesday	3024
3	Monday	2794
4	Sunday	2624
5	Friday	3538
6	Thursday	3239
7	Tuesday	2973

8. Monthly Trends form Total Orders

– sort by month –

```
select
    DATENAME(month, order_date) as month_name,
    count(distinct order_id) as total_orders,
    month(order_date) as number from pizza_sales
group by
    DATENAME(month, order_date),
    month(order_date)
order by number
```

	month_name	total_orders	number
1	January	1845	1
2	February	1685	2
3	March	1840	3
4	April	1799	4
5	May	1853	5
6	June	1773	6
7	July	1935	7
8	August	1841	8
9	September	1661	9
10	October	1646	10
11	November	1792	11
12	December	1680	12

– sort by descending or desc (penjualan terbesar ke kecil) –

```
select
    DATENAME(month, order_date) as month_name,
    count(distinct order_id) as total_orders
from pizza_sales
group by
    DATENAME(month, order_date)
order by total_orders desc
```

	month_name	total_orders
1	July	1935
2	May	1853
3	January	1845
4	August	1841
5	March	1840
6	April	1799
7	November	1792
8	June	1773
9	February	1685
10	December	1680
11	September	1661
12	October	1646

9. Trends by Pizza Category

```
select pizza_category,  
cast(sum(total_price) as decimal(10,2)) as total_sales,  
cast(sum(total_price) * 100 / (select sum(total_price) from pizza_sales) as decimal(10,2))  
as percentage_of_sales  
from pizza_sales  
group by pizza_category  
order by percentage_of_sales desc
```

	pizza_category	total_sales	percentage_of_sales
1	Classic	220053.10	26.91
2	Supreme	208197.00	25.46
3	Chicken	195919.50	23.96
4	Veggie	193690.45	23.68

10. Trends by Pizza Size

```
select pizza_size,  
cast(sum(total_price) as decimal(10,2)) as total_sales,  
cast(sum(total_price) * 100 / (select sum(total_price) from pizza_sales) as decimal(10,2))  
as percentage_of_sales  
from pizza_sales  
group by pizza_size  
order by percentage_of_sales desc
```

	pizza_size	total_sales	percentage_of_sales
1	L	375318.70	45.89
2	M	249382.25	30.49
3	S	178076.50	21.77
4	XL	14076.00	1.72
5	XXL	1006.60	0.12

11. TOP 5 Pizzas by Revenue

```
select TOP 5 pizza_name, cast(sum(total_price) as decimal(10,2)) as total_revenue from pizza_sales  
group by pizza_name  
order by total_revenue desc
```

	pizza_name	total_revenue
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Pizza	42768.00
3	The California Chicken Pizza	41409.50
4	The Classic Deluxe Pizza	38180.50
5	The Spicy Italian Pizza	34831.25

12. BOTTOM 5 Pizzas by Revenue

```
select top 5 pizza_name, cast(sum(total_price) as decimal(10,2)) as total_revenue from pizza_sales  
group by pizza_name  
order by total_revenue asc
```

	pizza_name	total_revenue
1	The Brie Carre Pizza	11588.50
2	The Green Garden Pizza	13955.75
3	The Spinach Supreme Pizza	15277.75
4	The Mediterranean Pizza	15360.50
5	The Spinach Pesto Pizza	15596.00

13. Most Pair Pizza (Cross Selling Opportunities)

a. Create Pizza Pair View

```
CREATE VIEW pizza_mostPair AS
SELECT
    a.pizza_name AS pizza_a,
    b.pizza_name AS pizza_b,
    COUNT(DISTINCT a.order_id) AS pair_count
FROM dbo.pizza_sales AS a
INNER JOIN dbo.pizza_sales AS b
    ON a.order_id = b.order_id
    AND a.pizza_name < b.pizza_name
GROUP BY a.pizza_name, b.pizza_name
HAVING COUNT(DISTINCT a.order_id) >= 10;
```

b. Bottom 5 Pizza Pair

```
SELECT TOP 5 *
FROM pizza_mostPair
ORDER BY pair_count ASC;
```

	pizza_a	pizza_b	pair_count
1	The Brie Carre Pizza	The Calabrese Pizza	17
2	The Brie Carre Pizza	The Chicken Alfredo Pizza	18
3	The Brie Carre Pizza	The Green Garden Pizza	21
4	The Brie Carre Pizza	The Napolitana Pizza	21
5	The Brie Carre Pizza	The Italian Vegetables Pizza	24

c. Top 5 Pizza Pair

```
SELECT TOP 5 *
FROM pizza_mostPair
ORDER BY pair_count Desc;
```

	pizza_a	pizza_b	pair_count
1	The Hawaiian Pizza	The Thai Chicken Pizza	270
2	The Barbecue Chicken Pizza	The Pepperoni Pizza	269
3	The Hawaiian Pizza	The Pepperoni Pizza	259
4	The Pepperoni Pizza	The Thai Chicken Pizza	257
5	The California Chicken Pizza	The Pepperoni Pizza	256

14. Most Ingredient Used & Least Ingredient Used

a. Create Table Ingredient

```
CREATE TABLE pizza_ingredients (  
    order_id INT,  
    order_date DATE,  
    pizza_name NVARCHAR(100),  
    ingredient NVARCHAR(100)  
);  
  
INSERT INTO pizza_ingredients (order_id, order_date, pizza_name, ingredient)  
SELECT  
    p.order_id,  
    p.order_date,  
    p.pizza_name,  
    LTRIM(RTRIM(value)) AS ingredient  
FROM pizza_sales p  
CROSS APPLY STRING_SPLIT(p.pizza_ingredients, ',');
```

	order_id	order_date	pizza_name	ingredient
1	1	2015-01-01	The Hawaiian Pizza	Sliced Ham
2	1	2015-01-01	The Hawaiian Pizza	Pineapple
3	1	2015-01-01	The Hawaiian Pizza	Mozzarella Cheese
4	2	2015-01-01	The Classic Deluxe Pizza	Pepperoni
5	2	2015-01-01	The Classic Deluxe Pizza	Mushrooms
6	2	2015-01-01	The Classic Deluxe Pizza	Red Onions
7	2	2015-01-01	The Classic Deluxe Pizza	Red Peppers
8	2	2015-01-01	The Classic Deluxe Pizza	Bacon
9	2	2015-01-01	The Five Cheese Pizza	Mozzarella Cheese
10	2	2015-01-01	The Five Cheese Pizza	Provolone Cheese
11	2	2015-01-01	The Five Cheese Pizza	Smoked Gouda Cheese
12	2	2015-01-01	The Five Cheese Pizza	Romano Cheese
13	2	2015-01-01	The Five Cheese Pizza	Blue Cheese
14	2	2015-01-01	The Five Cheese Pizza	Garlic
15	2	2015-01-01	The Italian Supreme Pizza	Calabrese Salami
16	2	2015-01-01	The Italian Supreme Pizza	Capocollo

b. Top 5 Most Ingredients Used

```
select top 5  
    ingredient,  
    COUNT(*) AS total_usage  
from pizza_ingredients  
group by ingredient  
order by total_usage desc;
```

	ingredient	total_usage
1	Garlic	27422
2	Tomatoes	26601
3	Red Onions	19547
4	Red Peppers	16284
5	Mozzarella Cheese	10333

c. Top 5 Least Ingredients Used

```
select top 5
    ingredient,
    COUNT(*) AS total_usage
from pizza_ingredients
group by ingredient
order by total_usage asc;
```

	ingredient	total_usage
1	Brie Carre Cheese	480
2	Thyme	480
3	Prosciutto	480
4	Pears	480
5	Caramelized Onions	480